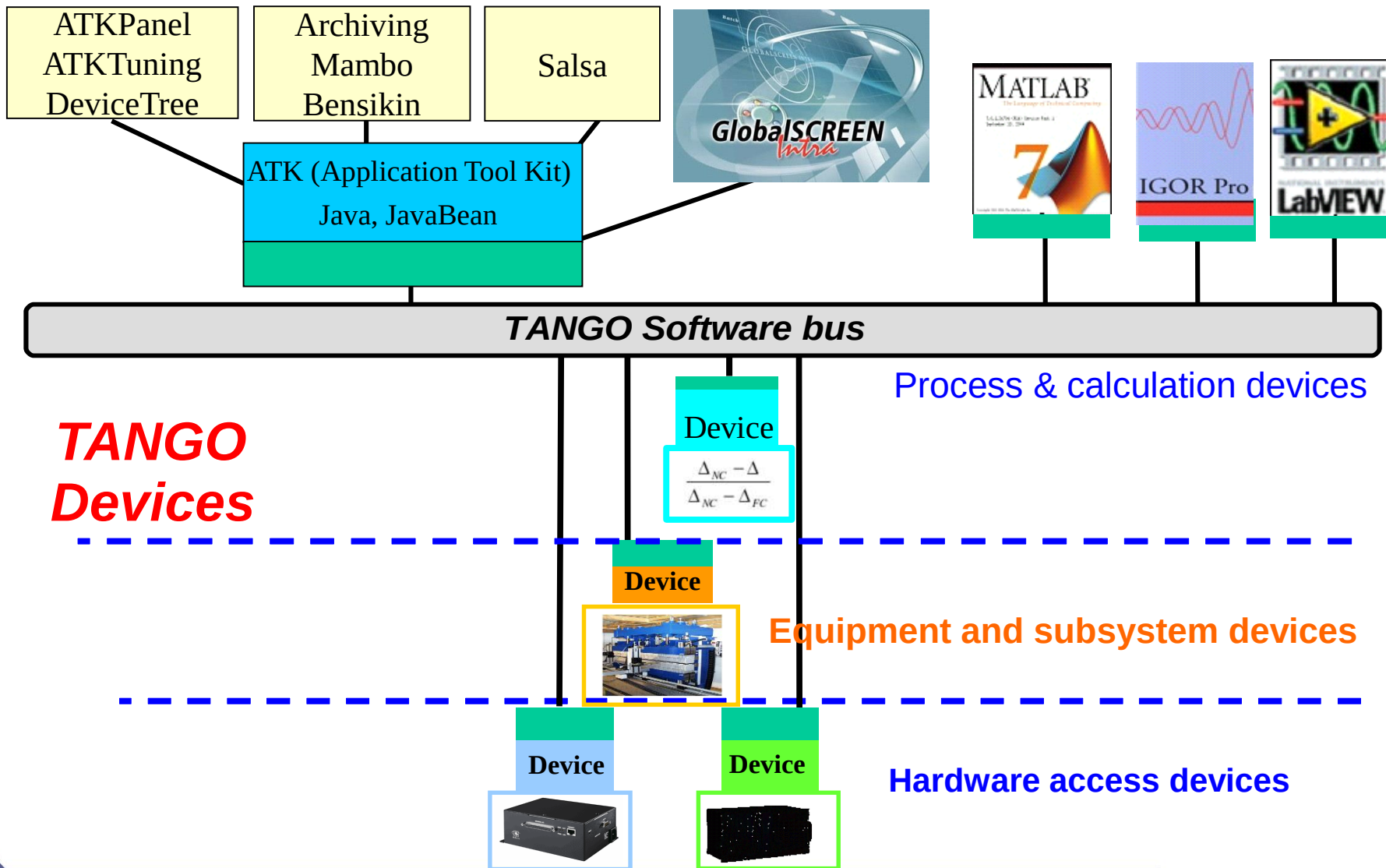


TANGO

Java Generic Devices development at SOLEIL

Katy Saintin

- Software architecture
- Why generic devices ?
- Presentation of each generic devices.
- Results.
- Questions.



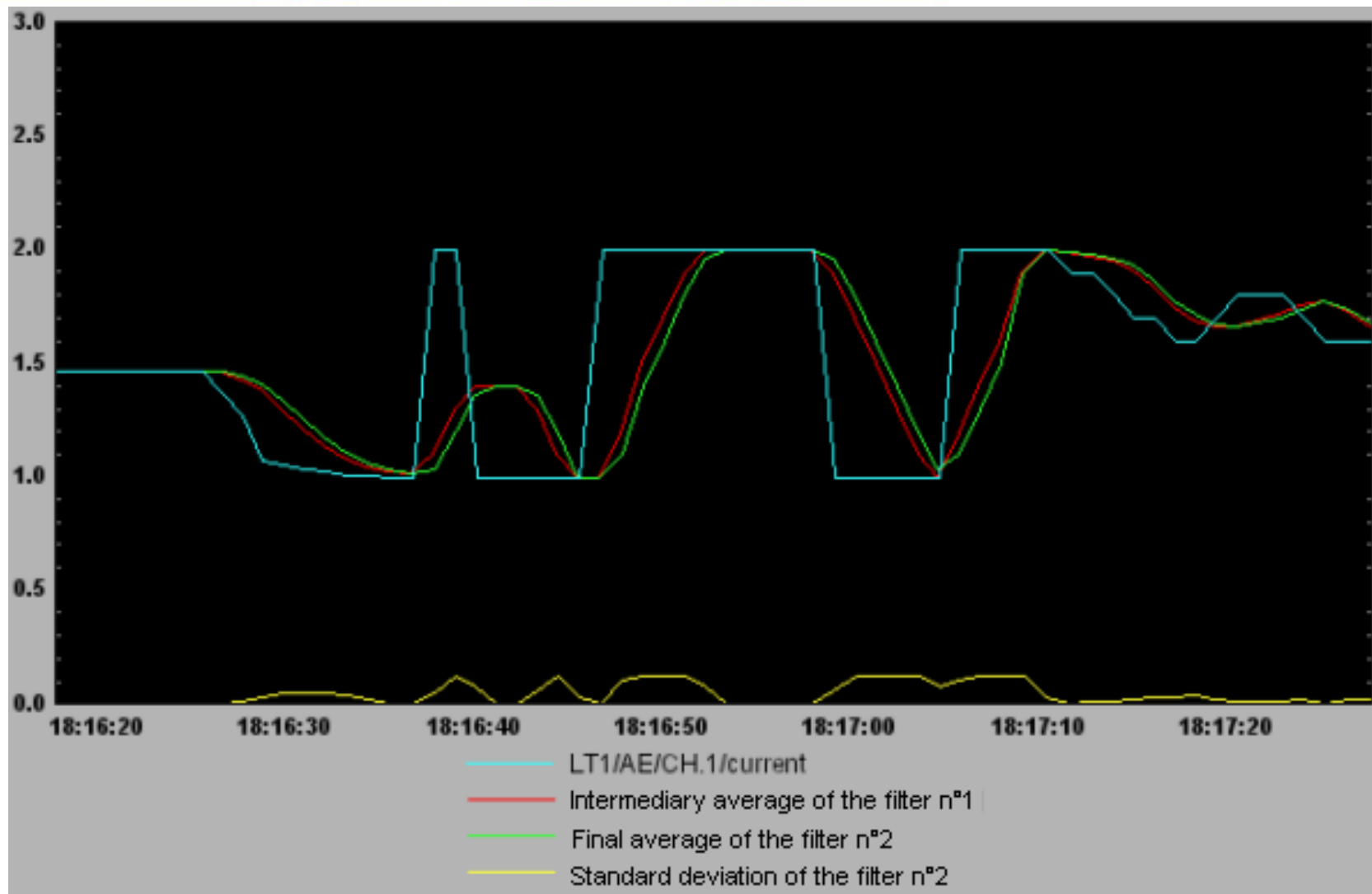
Why generic devices ?

- To keep all the intelligence in the device.
 - ✓ To make the GUI ready to use quickly.
 - ✓ To make the GUI easy to maintain.
 - ✓ To limit the number of iterations in the GUI development.
- To provide a modular solution.
 - ✓ To carry out a device server without programming.
 - ✓ To solve a recurring problem.
 - ✓ To make a device easy to maintain.
 - ✓ To limit the number of iterations in the device development.
- To isolate a complex interface into simple entities
 - ✓ To employ all the ready to use tools dedicated to these entities.
 - polling, archiving...

- Averager
- TangoParser
- BooleanAttributeExtractor
- EnumeratedAttribute
- EnumeratedCommand
- AttributeComposer
- StateComposer
- FacadeDevice

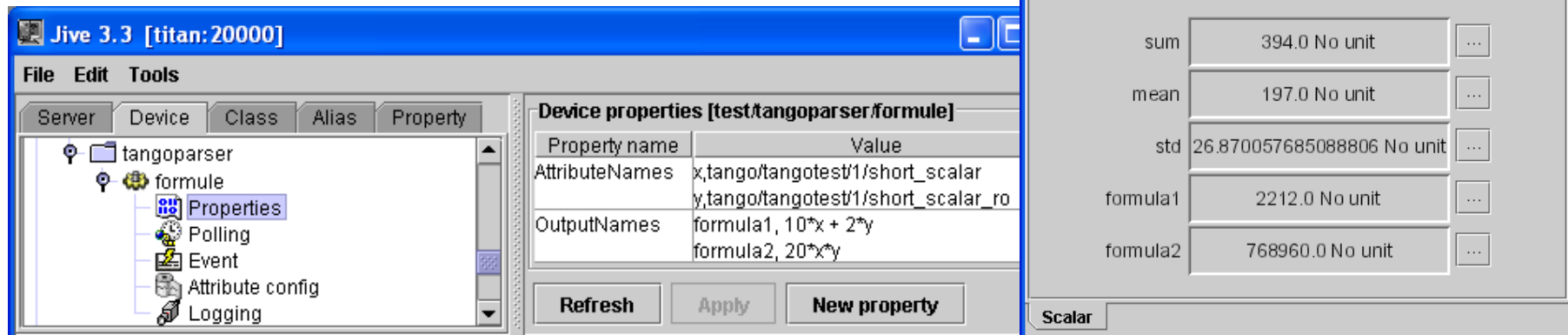
Calculation of the average and the standard deviation of any numerical tango scalar attribute :

- To follow a slow data variation.
- To archive less values.
- To smooth a curve.
- To eliminate the noise and to apply a filtering.



Calculation of tango attributes the values of which are mathematical expressions of other tango attributes. The expression can be defined at runtime in a mathematical syntax.

- To convert a data.
- To average several attributes.
- In short to make any type of calculations



The screenshot shows the Jive 3.3 [titan:20000] interface. On the left, a tree view shows the 'tangoparser' device with sub-items: 'formule', 'Properties', 'Polling', 'Event', 'Attribute config', and 'Logging'. The 'Properties' item is selected. The main panel displays 'Device properties [test/tangoparser/formule]' with a table of properties and values.

Property name	Value
AttributeNames	x,tango/tangotest1/short_scalar y,tango/tangotest1/short_scalar_ro
OutputNames	formula1, 10*x + 2*y formula2, 20*x*y

Below the table are buttons for 'Refresh', 'Apply', and 'New property'.

On the right, a separate window titled 'AtkPanel : test/tangoparser/formule' shows the status and results. The status indicates 'Device is ready to calculate values of: formula1 formula2'. Below this, a table shows calculated values for 'sum', 'mean', 'std', 'formula1', and 'formula2'.

Property	Value
sum	394.0 No unit
mean	197.0 No unit
std	26.870057685088806 No unit
formula1	2212.0 No unit
formula2	768960.0 No unit

At the bottom of the AtkPanel window, there is a 'Scalar' button.

Extraction of several Boolean values from any scalar attribute representing a field of bytes.

- To poll the Boolean attribute.
- To realized a simple GUI with an existing ATK widget.
- To archive a simple information.

Attribute property editor

Identification

Device D09-1-C02/MVS.1

Attribute combinedState

23229.0 No unit

Properties

Label Combined State

Minimum value Not specified **Maximum value** Not specified

Minimum alarm Not specified **Maximum alarm** Not specified

Format No format **Unit** No unit

combined boolean states of the valve
 // bit 0 : BSV_VS_Ouverte Valve opened
 // bit 1 : BSV_VS_Fermee Valve Closed
 // bit 2 : BSV_VS_Default Valve faulty
 // bit 3 : BSV_VS_Mouvement valve moving
 // bit 4 : BSV_VS_Securisee valve securized
 // bit 5 : BSV_VS_Default_Memorise memorized fault state
 // bit 6 : BSV_Interlocked valve closed on external interlock

Apply change Information Dismiss

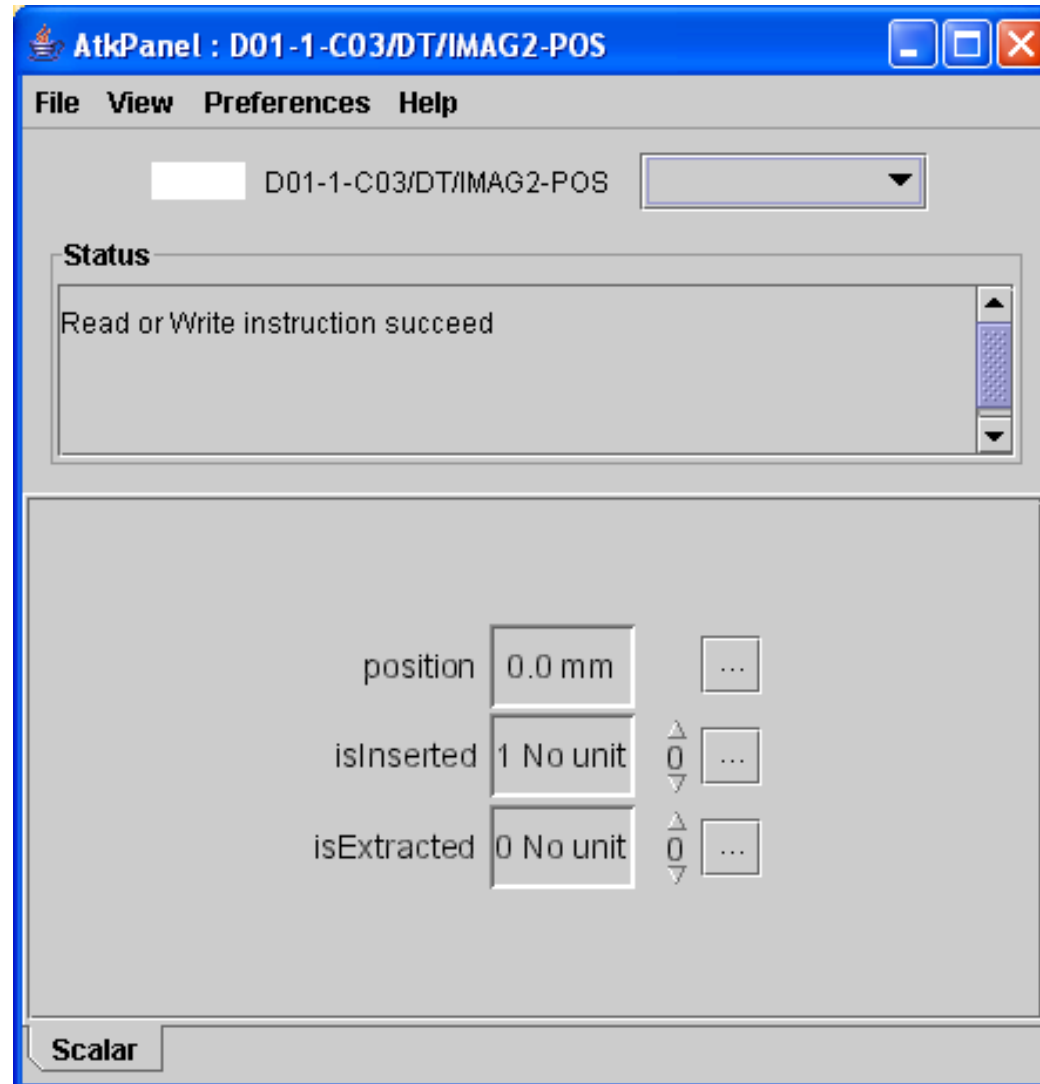
combinedState	19380.0 No unit	...
binaryCode	100101110110100	...
isOpen	0 No unit	...
isClosed	0 No unit	...
isDefault	1 No unit	...
isMoving	0 No unit	...
isSecurized	1 No unit	...
isDefaultMemorized	1 No unit	...
isInterlocked	0 No unit	...

To predefine an enumerated list of values for a tango scalar attribute. Boolean attributes are dynamically created.

- To poll the Boolean attribute.
- To move the intelligence of the ATK widget in the device.
- To carry out a simple GUI with a simple ATK widget.
- To archive a simple information.
- To give an explicit name to an attribute value.

 [See device documentation](#)

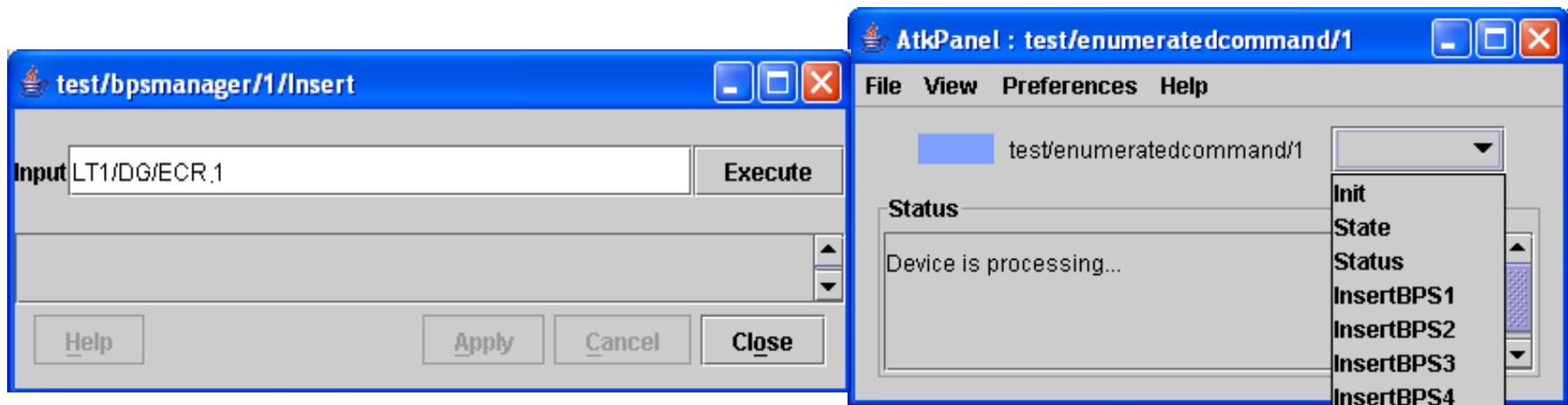
EnumeratedAttribute (java)



To predefine an enumerated list of values for a tango scalar command. Void command are dynamically created.

- To poll a void command.
- To move the intelligence of the ATK widget in the device.
- To realized a simple GUI with a simple ATK widget.
- To give explicit names to a command value.

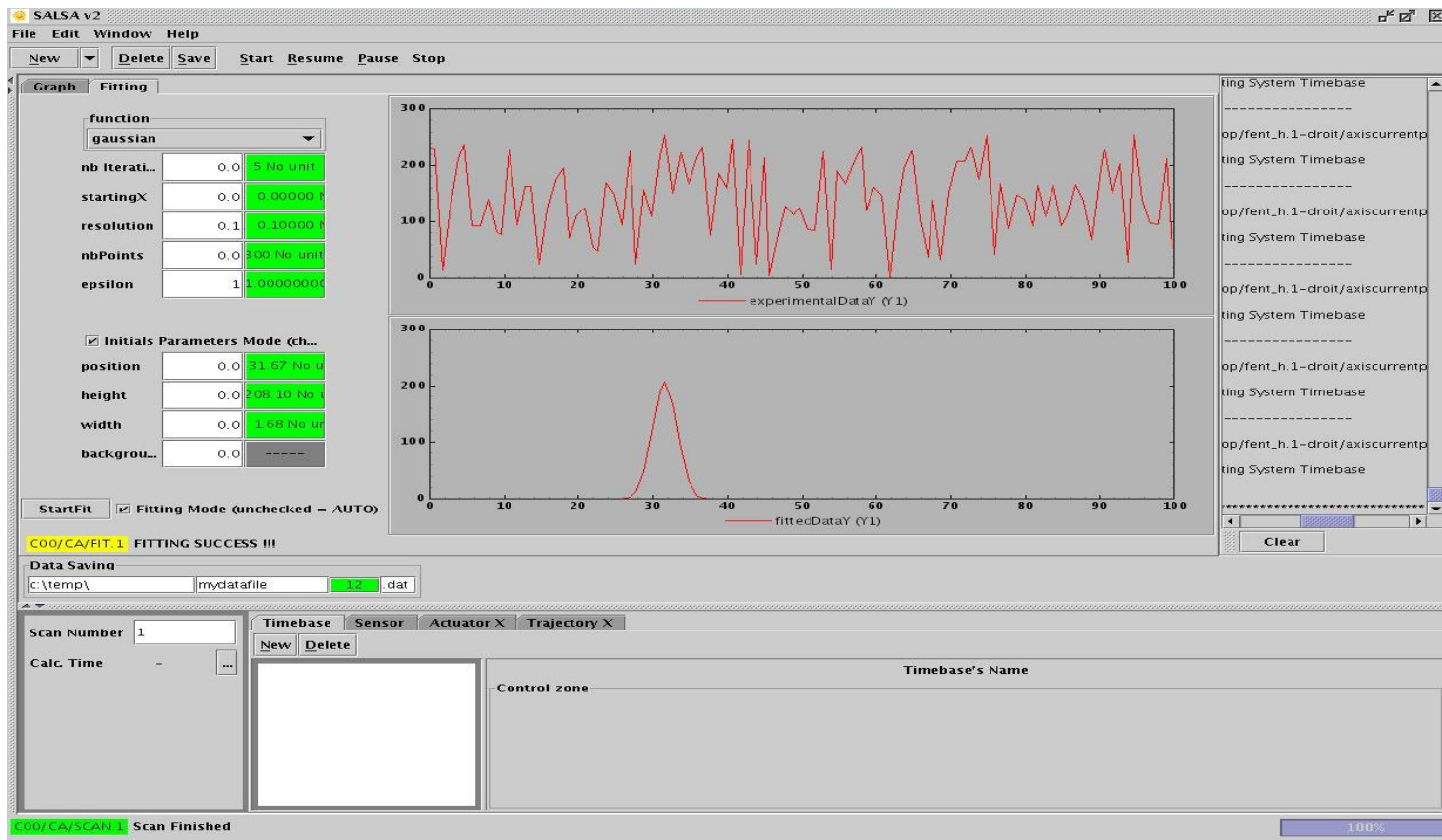
[!\[\]\(b3131996c2d47980618867ba93d92313_img.jpg\) See device documentation](#)



To extract a simple scalar data from a spectrum or a spectrum from an image tango attribute.

- To handle multidimensional data (2D,3D) projection.
- To archive this data.
- To represent this data with a existing ATK widget.

To apply a Gaussian or Lorentzian function, in order to get different pieces of information like the width of the peak, the half height intensity....

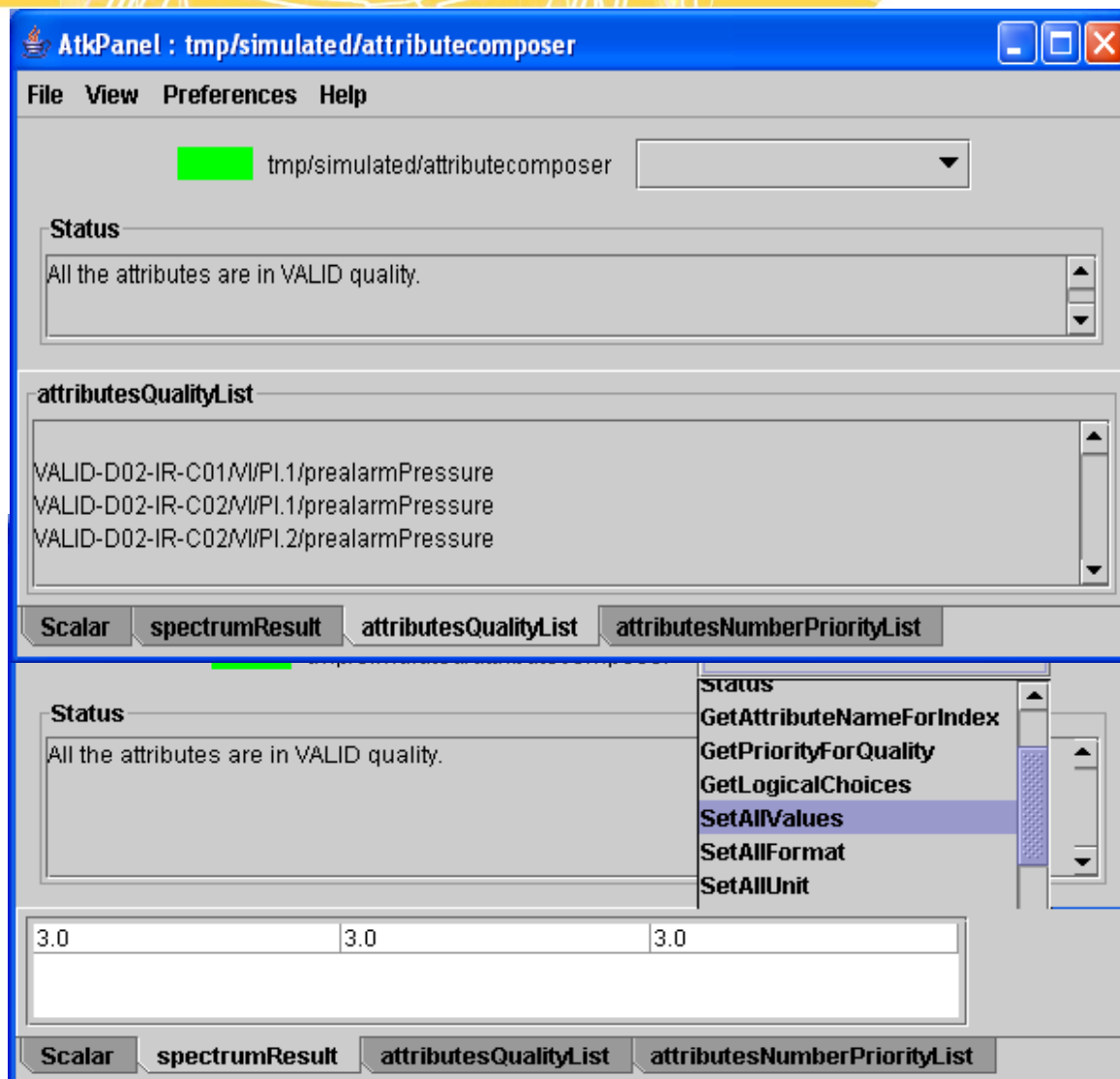


To build a spectrum attribute starting from simple scalar tango attributes.

- To make a soft synchronization of the data.
- To archive a set of data.
- To display easily all the data in a chart.
- To deduce a general state according to the attribute quality.
- To represent the general state with a simple ATK widget.
- To write the same value on several attribute.
- To give the logical result (OR, XOR, AND gate), on a list of Boolean attributes.

 [See device documentation](#)

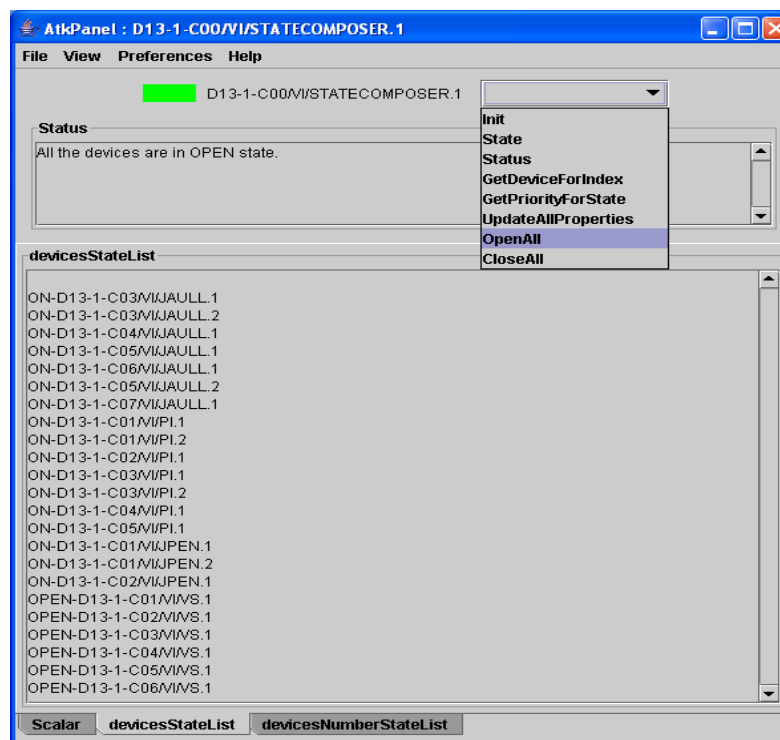
AttributeComposer (java)



To deduce general state according to a list of devices.

- To archive the state information.
- To represent the general state with a simple ATK widget.
- To execute a grouped command on several device.

☞ [See device documentation](#)



Construction of a new device starting from attributes, commands and state of other devices.

- To hide a part of a device.
- To install quickly a simulated device in order to test a GUI.
- To gather information in only one device.
- To create a basic device server without developing.
- To create an intermediary device when the tango interface of a device changes.

 [See device documentation](#)

FacadeDevice (java)

Device properties [tmp/simulated/mirror]

Property name	Value
AttributeLabelList	tx tz ty
AttributeNameList	D13-1-C02/OP/MIR1-MT_TZ1/position D13-1-C02/OP/MIR1-MT_TZ2/position D13-1-C02/OP/MIR1-MT_TZ3/position
CommandLabelList	Stop
CommandNameList	D13-1-C02/OP/MI1-STATECOMPOSER/StopAll
DeviceNameState	D13-1-C02/OP/MI1-STATECOMPOSER

AtkPanel : tmp/simulated/mirror

File View Preferences Help

tmp/simulated/mirror

Status

All the devices are in STANDBY state.

Init
State
Status
Stop

tx 1.0 mm 000.00 ...

tz 2.0 mm 002.00 ...

ty 10.0 mm 000.00 ...

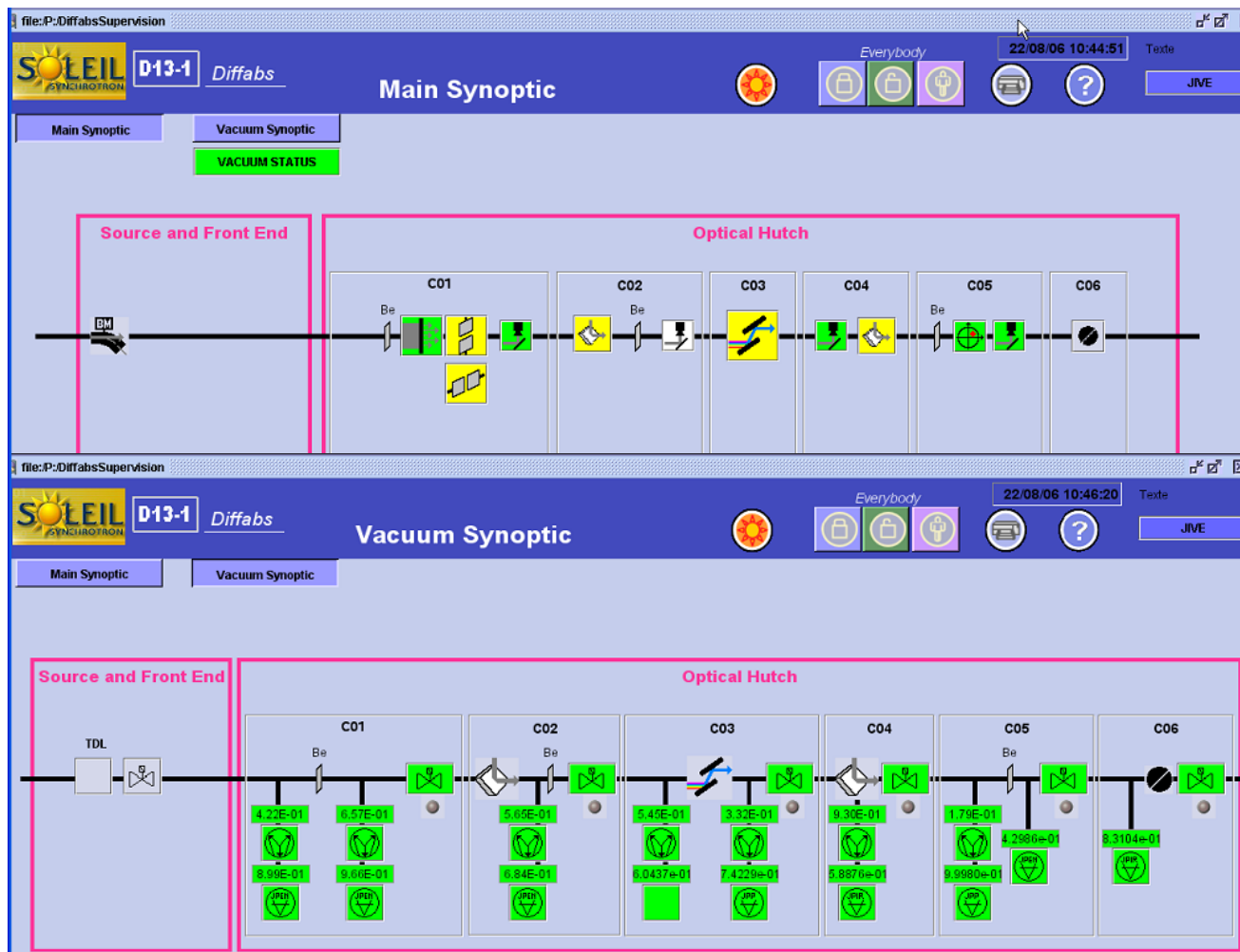
Scalar unknownAttributesList unknownCommandsList

➤ New device servers :

- ✓ Supervision of a group of devices (Composers)
 - ❑ Group of power supplies, ion pumps, vacuum valve...
- ✓ Installation of a simulated tango environment (FacadeDevice)
 - ❑ Adapting TangoTest to any required DeviceServer interface
- ✓ Devices providing a near hardware solution
 - ❑ Motorized Insertion/Extraction in the beam.
 - ❑ Uncoupling motorized device like a mirror.
 - ❑ Extraction of a device default information.

➤ GUI development.

- ✓ ATKPanel answers quickly to a supervision problem.
- ✓ Specific GUIs are easily developed.



- Boolean attributes
- Deletion of dynamics attributes and commands
- Group and asynchronies commands
- Memorized attributes.

Any questions ?

